Zin Win Thu, Ph.D.

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Education

Konkuk University — Seoul, South Korea
M.Sc.-Ph.D. in Aerospace Information Engineering — *Mar. 2020 – Feb. 2026*

- Dissertation: *GenAI Optimization Approach for Large Scale Aircraft Design under Uncertainty*.
- Supervisor: Prof. Jae-Woo Lee.

Myanmar Aerospace Engineering University — Meiktila, Myanmar
B.E. in Aerospace Engineering — *Mar. 2012 – Dec. 2018*

- Major in Propulsion and Flight Vehicles.
- GPA: 4.9/5.

Employment History

Research Assistant Professor — Mar. 2026 – Present
Konkuk Aerospace Design and Airworthiness Institute, Konkuk University — *Seoul, South Korea*

- Conduct research on uncertainty-aware 3D aircraft design optimization integrating AI methods.
- Lead the development of the in-house Comprehensive Aircraft Design Environment (CADE) across M.S. and Ph.D. students, research assistants, and senior researchers.
- Guide and supervise graduate students in research projects and publications.

Graduate Researcher — Mar. 2020 – Feb. 2026
Konkuk Aerospace Design and Airworthiness Institute, Konkuk University — *Seoul, South Korea*

- Contributed to institutional research projects related to UAV, UAM, RAM design, and digital twin technologies.
- Conducted research in multidisciplinary aircraft design optimization, sizing, and performance analysis, with emphasis on electric and hybrid-electric aircraft.
- Investigated machine learning and generative AI approaches for aircraft geometry representation, aerodynamic prediction, and uncertainty quantification in conceptual design.

Lead Developer, CADE Aircraft Design Environment — 2024 – Present
Konkuk Aerospace Design and Airworthiness Institute, Konkuk University — *Seoul, South Korea*

- Lead the development of the in-house Comprehensive Aircraft Design Environment (CADE) for conceptual and preliminary aircraft design for versatile aircraft
- Coordinate development tasks across M.S. and Ph.D. students, research assistants, and senior researchers.
- Integrate analysis modules, design workflows, optimization algorithms, and visualization tools into a unified aircraft design framework.

Research Intern — 2016 – 2018
Fluid Energy and Environmental Engineering Lab, Myanmar Aerospace Engineering University — *Meiktila, Myanmar*

- Conducted research on multidisciplinary design analysis of fixed-wing aircraft.
- Gained early experience in conceptual aircraft design.
- Contributed to journal and conference publications.

Journal Publications

- **Zin Win Thu**, Aye Aye Maw, and Jae-Woo Lee, "Deep Generative and Probabilistic Surrogate Modeling for Subsonic Airfoil Design," *Journal of Aircraft*, under review, 2025.
- Min Ji Kim, Mohammad Irfan Alam, **Zin Win Thu**, Dong Oh Kwang, and Jae-Woo Lee, "Multi-step Hybrid Optimization Strategy with Adaptive Boundary Control Case Study: Advanced Air Vehicle Wing Design Optimization," *Aerospace Science and Technology*, 2026, Article 111979.

- **Zin Win Thu**, Maxim Tyan, Yong-Hyeon Choi, Mohammad Irfan Alam, and Jae-Woo Lee, "Possibility-Based Sizing Method for Hybrid Electric Aircraft," *IEEE Access*, 2025.
- Yong-Hyeon Choi, **Zin Win Thu**, Maxim Tyan, Woosuk Jung, and Jae-Woo Lee, "Balancing Power and Energy Needs in Hybrid Electric Aircraft Sizing: Payload and Technology Considerations," *Journal of the Korean Society for Aeronautical & Space Sciences*, vol. 53, no. 4, pp. 421–432, 2025.
- Mohammad Irfan Alam, **Zin Win Thu**, Nghia Nguyen, Maxim Tyan, and Jae-Woo Lee, "Integrated Propulsion System Analysis Framework for Designing Advanced Air Mobility Aircraft," *IEEE Transactions on Transportation Electrification*, vol. 10, no. 4, pp. 10219–10238, 2024.
- **Zin Win Thu**, Dasom Kim, Junseok Lee, Woon-Jae Won, Hyeon Jun Lee, Nan Lao Ywet, Aye Aye Maw, and Jae-Woo Lee, "Multivehicle Point-to-Point Network Problem Formulation for UAM Operation Management Used with Dynamic Scheduling," *Applied Sciences*, vol. 12, no. 22, Article 11858, 2022.
- **Zin Win Thu**, Aung Myo Thu, and Phyo Wai Thaw, "Design and Optimization of Aircraft Configuration for Minimum Drag," *International Journal of Aviation, Aeronautics, and Aerospace*, vol. 5, no. 5, Article 10, 2018.

Selected Conference Papers

- **Zin Win Thu**, Min Ji Kim, and Jae-Woo Lee, "Efficient Aircraft Design through Deep Generative Modeling," ACSMO 2026, Busan, Korea, 2026.
- Yong-Hyeon Choi, **Zin Win Thu**, Dongoh Kwag, and Jae-Woo Lee, "Deriving the Flight Envelope and Conversion Corridor of an eVTOL Tiltrotor Aircraft Considering Battery Discharge Rate," AIAA AVIATION Forum and ASCEND, 2025, p. 3758.
- DongOh Kwang, Min Ji Kim, **Zin Win Thu**, and Jae-Woo Lee, "A multi-objective optimization approach for reliable electric propulsion system architecture in AAM aircraft," WCSMO16, Kobe, Japan, 2025.
- **Zin Win Thu**, Jae-Hyun Ahn, Jae-Lyun Lee, Do-Youn Kwon, Yeon-Ju Choi, Woon-Jae Won, Maxim Tyan, and Jae-Woo Lee, "Enhanced Performance Prediction of Hydrogen Fuel Cell Powered eVTOL UAV," AIAA AVIATION 2022 Forum, 2022, p. 3382.

Research Project Experience

Development of Comprehensive Aircraft Design Environment (CADE) 2022 – Present
KADA, Konkuk University *Seoul, South Korea*

- Serve as team lead for the development of an integrated aircraft design and analysis environment.
- Develop modules for aerodynamic analysis, propeller analysis, aircraft performance evaluation, and design optimization.
- Support the integration of analysis tools into a unified framework for efficient conceptual and preliminary aircraft design workflows.

Concept Development of Next-Generation Regional/Urban Air Mobility Aircraft 2023 – 2024
KADA, Konkuk University *Seoul, South Korea*

- Developed an electric aircraft sizing program for regional and urban air mobility concepts.
- Built design evaluation tools to assess performance and mission capability.
- Delivered a seminar to Hyundai Motor Company on aircraft sizing.

UAM Operational Digital Twin 2021 – 2023
KADA, Konkuk University *Seoul, South Korea*

- Contributed to digital twin research for urban air mobility mission operations.
- Developed methodologies for multi-vehicle mission scheduling in operational environments.
- Supported mission planning for UAM operations using path planning methods such as A* and ADA*.

Development of Small eVTOL UAV with Hydrogen Fuel Cell 2020 – 2023
KADA, Konkuk University *Seoul, South Korea*

- Performed performance analysis for a small eVTOL UAV powered by a hydrogen fuel cell system.
- Conducted wind tunnel testing for propellers including electric propulsion system.

Development of Personal Air Vehicle (PAV) 2020 – 2021
KADA, Konkuk University *Seoul, South Korea*

- Participated in the conceptual design and analysis of personal air vehicle configurations.
- Contributed to aircraft sizing and performance analysis for a hydrogen fuel cell powered personal air vehicle.

Professional Distinctions

- **Professor's Scholarship**, Konkuk University, 2020 – 2025.
- **Outstanding Researcher Grant**, Konkuk University Research Fund, 2024.
- **Mentorship Award**, Research for Undergraduate Students, Konkuk University, 2024.
- **Granted Patent**, Korean Patent No. 10-2023-0036070, South Korea, 2023.
- **Project Funding Grant**, Design and Construction of Mini-Jet Engine, Myanmar Aerospace Engineering University, 2018.

Teaching and Mentoring

Teaching Assistant, Comprehensive Aircraft Design Class 2023 – 2024
Department of Aerospace Information Engineering, Konkuk University *Seoul, South Korea*

- Supported the instruction of a senior-level aircraft design course with 34 undergraduate students.
- Mentored team projects in aircraft sizing, aerodynamics, propulsion, and performance analysis.
- Delivered invited lectures on electric aircraft sizing and aircraft performance analysis.

Research Mentor 2023
Research for Undergraduate Students, Konkuk University *Seoul, South Korea*

- Mentored an undergraduate researcher in electric regional aircraft sizing and powertrain comparison.
- Supervised the development of computational modules for aircraft sizing and conceptual design analysis.

Languages

Languages : English, fluent professional proficiency; Burmese, native; Korean, basic

References

Available upon request.